

A Panel-Contrast Nuisance Basis: A Finite Repair and Weighting Obstruction

Liang Wang

School of Mathematics and Statistics, Huazhong University of Science and Technology
Wuhan, China

2 September 2026

Abstract

The preceding finite cross-panel audit found that one nuisance coefficient vector does not fit two protocol-compatible twin-prime response panels. We test the smallest structured relaxation that still shares coefficients within each panel: one signed panel-contrast column is added for each of three nuisance categories. The resulting six-column basis is exactly equivalent to one shared nuisance vector per panel. On the locked six-row finite panel, its raw-energy residual retention is 0.2962189247, just below the inherited 0.30 guard. The same basis retains 0.3186506700 after equal-row normalization, and four cross-fit predictions retain 0.3759486734–0.6342934197. Thus the raw crossing is a finite, weighting-sensitive partial repair rather than a stable canonical law. The 18 contrast holdout records retain 0.6372238668–0.9128543547. No arithmetic, asymptotic, Route-A/Route-B, or twin-prime conclusion follows.

1 Question and scope

TPC-343 showed that a single nuisance coefficient vector across its TPC-341 and TPC-342 panels fails an in-sample residual guard under both raw and equal-row weighting. The next minimal question is whether the mismatch is explained by a panel-level contrast while retaining a common law inside each panel. This paper makes that relaxation explicit and audits it independently.

All objects below are finite vectors generated by the repository’s locked source/operator model. The words “twin” and “nuisance” name masks and response vectors; they do not assert an arithmetic noise decomposition. In particular, a finite fit is not a source-uniform estimate and does not pay any Route-B loss budget.

2 Frozen finite protocol

The two panels are inherited without modification:

$$\begin{aligned}\mathcal{P}_{341} &: [48097, 48608], [48609, 49120], [49217, 49728], \\ \mathcal{P}_{342} &: [40097, 40608], [40609, 41120], [41121, 41632].\end{aligned}$$

Each row has source length 512 and scale 1024. The operator is the all-plus deleted-diagonal operator with $Q = 54$, kernel exponent one, and height $H = 66$. The nine controls are the identity, reversal, and seven fixed affine coordinate bijections. There are four source masks: twin-prime, non-twin-prime shift, prime-power shift, and zero support.

For row r , write $y_r \in \mathbb{R}^{512}$ for the nine-control mean of the twin output and $n_{r,j} \in \mathbb{R}^{512}$ for the mean of nuisance category j , where j ranges over the three nuisance categories. The six rows are ordered with the three TPC-341 rows first. Define

$$b_j = (n_{1,j}, n_{2,j}, n_{3,j}, n_{4,j}, n_{5,j}, n_{6,j})^T, \quad d_j = (n_{1,j}, n_{2,j}, n_{3,j}, -n_{4,j}, -n_{5,j}, -n_{6,j})^T.$$

The sign vector $(+1, -1)$ is fixed before the numerical readout. The baseline shared model uses the three columns b_j ; the contrast model uses the six columns b_j, d_j .

For equal-row weighting, each y_r and all three $n_{r,j}$ in that row are divided by $\|y_r\|_2$. This is a declared sensitivity transform, not a canonical normalization. The inherited in-sample decision guard is residual retention < 0.30 ; the hostile leave-one-control-out guard is retention > 0.40 .

3 Exact finite structure

Proposition 1 (contrast reparameterization). *For every nuisance category j , let*

$$u_{1,j} = \frac{b_j + d_j}{2}, \quad u_{2,j} = \frac{b_j - d_j}{2}.$$

Then $u_{1,j}$ is supported on the TPC-341 block, $u_{2,j}$ is supported on the TPC-342 block, and

$$\text{span}\{b_j, d_j\} = \text{span}\{u_{1,j}, u_{2,j}\}.$$

Proof. The definitions give

$$u_{1,j} = (n_{1,j}, n_{2,j}, n_{3,j}, 0, 0, 0)^T, \quad u_{2,j} = (0, 0, 0, n_{4,j}, n_{5,j}, n_{6,j})^T.$$

Thus both $u_{1,j}$ and $u_{2,j}$ are linear combinations of b_j, d_j . Conversely $b_j = u_{1,j} + u_{2,j}$ and $d_j = u_{1,j} - u_{2,j}$. Each span contains the other. $\square$

The proposition is a finite change of coordinates. It says exactly what the new model permits: separate nuisance coefficients for the two panels, with the coefficient vector still shared over the three rows inside each panel. It does not say that the two coefficient vectors are canonical or persist for new source windows.

For any finite nuisance matrix N , let P_N be the Euclidean orthogonal projector onto its column space. We use the exact identity

$$\|Y\|_2^2 = \|P_N Y\|_2^2 + \|(I - P_N)Y\|_2^2. \quad (1)$$

The implementation constructs P_N from the positive singular vectors and checks the decomposition gap. A rational anchor, independent of the source calculation, uses $b = (1, 0, 0, 1)^T$, $d = (1, 0, 0, -1)^T$. The two reparameterized columns are $u_1 = (1, 0, 0, 0)^T$ and $u_2 = (0, 0, 0, 1)^T$; projecting $Y = (1, 1, 1, 1)^T$ has energy 2 and residual energy 2.

4 Audited finite readout

The producer creates $2 \cdot 3 \cdot 9 \cdot 4 = 216$ raw records, of which 171 are nonempty. There are six in-sample row means, 18 contrast holdouts (nine omitted controls under each weighting), and four directional cross-fits. Table 1 gives the pooled readout.

The raw contrast result is a narrow crossing: its margin below 0.30 is approximately 0.0037810753. Equal-row weighting removes the crossing. The observed rank is five rather than six because the

Table 1: Pooled finite projections and transfer diagnostics.

quantity	value
row-block raw residual retention	0.2325429101
one-vector shared raw residual retention	0.3198013104
panel-contrast raw residual retention	0.2962189247
panel-contrast equal-row retention	0.3186506700
contrast positive rank (six columns declared)	5
contrast raw positive-condition diagnostic	141.9850
contrast holdout range, both weightings	0.6372238668–0.9128543547
cross-fit prediction range	0.3759486734–0.6342934197

prime-power nuisance column is zero on the three TPC-342 rows; the degeneracy is part of the declared data and is not discarded.

The exact span identity in Proposition 1 was also evaluated by constructing the equivalent panel-supported columns directly. The raw and equal-row residual retentions agree with the contrast construction to the declared numerical tolerance. This validates the reparameterization but does not turn the model into a universal basis.

5 Holdout and cross-fit tests

For each omitted control, nuisance columns are formed from the other eight controls, and the six held-out twin outputs are projected onto the contrast span. Across the nine omissions, the raw retention range is 0.6372238668–0.9039334258 and the equal-row range is 0.6880962075–0.9128543547. Hence all 18 hostile holdout values exceed the 0.40 guard. This is a finite out-of-sample diagnostic for the declared control orbit, not a probability statement.

As a separate transfer test, least-squares coefficients are fitted to the three row means of one panel and applied without refitting to the other. The four prediction retentions are

$$0.5156705854, \quad 0.3759486734 \quad (\text{raw, two directions})$$

and

$$0.6342934197, \quad 0.4261160161 \quad (\text{equal-row, two directions}).$$

All four exceed 0.30, so the low-residual cross-panel transfer criterion is refuted on this finite pair. These are prediction residuals, not orthogonal projection residuals; equation (1) is not asserted for them.

6 Independent implementation and claim firewall

The producer checks the TPC-343 parent certificate and then uses the hash-locked TPC-340 source and operator implementation. The independent checker does not import the producer: it uses the separately hash-locked reverse-shell engine, rebuilds the source masks and all nine controls, and reimplements the projection, contrast, holdout, and cross-fit calculations. A nine-mutation stress suite rejects changes to row geometry, panel signs, guard values, provenance semantics, rank, and the exact anchor. Normal and optimized executions are required to have empty standard error and byte-identical standard output.

Table 2: Claim ledger.

item	status
contrast span identity	proved exact finite declared model
raw contrast guard	numerically certified finite scoped pass
equal-row contrast guard	refuted scoped
cross-fit low-residual transfer	refuted scoped
source-uniform arithmetic L^2	open
uniform masked operator bound	open
full Route-B Gate B	open
fixed-power credit	0
twin-prime conclusion	none

The Session-named Route-A and Route-B evaluator files are absent in this checkout. The local Bridge-B script is consequently a fail-closed fallback, not an official evaluator pass. In particular, the finite raw crossing does not pay the strict $1/400$ endpoint and cannot be promoted to an asymptotic statement.

7 Conclusion and next question

The panel-contrast basis gives a useful but sharply delimited result. It explains enough of the cross-panel mismatch to move the raw pooled residual from 0.3198 to 0.2962, while its equal-row value 0.3187 and the cross-fit diagnostics prevent a robustness claim. The reusable structure is

shared span $\longrightarrow$ signed contrast $\longrightarrow$ panel-adaptive span $\longrightarrow$ weight and transfer audit.

The next natural finite question is geometric: compute principal angles between the two panel nuisance subspaces and test whether the apparent improvement is a stable Grassmannian relation or a coordinate/weighting artifact. Until a source-uniform arithmetic theorem is supplied, all Route-B and twin-prime gates remain open.